

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**End Semester Examination – Winter 2018****Course: B. Tech in Civil Engineering****Sem: III****Subject Name: Hydraulics I****Subject Code: BTCVC303****Max Marks: 60****Date: 05-12-2018****Duration: 3 Hr.****Instructions to the Students:**

1. Solve **all** questions out of the following.
2. The level question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q.1 Solve Any Three of the following.		
A) Explain the different applications of Fluid Mechanics in Civil Engineering.	CO2	04
B) Define: Surface Tension and Bulk modulus. Give proper unit of each.	CO2	04
C) Calculate the dynamic viscosity of oil, which is used for lubrication between a square plate of size 0.8 x 0.8 m ² and plane with angle of inclination 30°. The weight of square plate is 294.5 N and it slides down the inclined plane with uniform velocity of 0.3m/s. The thickness of an oil film is 1.5mm.	CO2	04
D) When a pressure of 20.7MN/m ² is applied to 100 liters of a liquid, it's volume decreases by 1 liter. Find the bulk modulus of the liquid and compressibility of liquid.	CO2	04
Q.2 Solve Any Two of the following.		
A) Explain in detail with neat sketch – i) Differential U-Tube Manometer ii) Bourdon Tube Pressure Gauge	CO2	06
B) Prove that the center of pressure of a vertical plane surface is always below the center of gravity.	CO3	06
C) A wooden block of relative density 0.7 has width 15cm, depth 30cm and length 150cm. it floats horizontally on the surface of water. Calculate the volume of water displaced, depth of immersion and position of center of buoyancy. Also find the metacentric height.	CO3	06
Q.3 Solve the following.		
A) Derive a three dimensional general continuity equation in Cartesian co-ordinates.	CO3	06
B) The velocity of component in a two dimensional flow are $u = 2xy \quad v = b^2 + x^2 - y^2$ i) Is the flow possible? ii) Determine potential function. iii) Determine corresponding stream function.	CO3	06

Q.4 Solve Any Two of the following

- A) Derive an expression for point velocity distribution for laminar flow between two fixed parallel plates. **CO3 06**
- B) Explain with neat sketches the difference between hydro dynamically smooth & rough boundaries. **CO2 06**
- C) Explain Prandtl's Mixing Length Theory and Nikuradse's experiment. **CO2 06**

Q. 5 Solve Any Two of the following.

- A) Write short note on Non-dimensional numbers – i) Froude number ii) Weber Number iii) Reynold's Number. **CO3 06**
- B) Explain concept of equivalent pipe & Derive Dupit's equation in form **CO3 06**
- $$\frac{L}{D^5} = \frac{L_1}{D_1^5} + \frac{L_2}{D_2^5} + \frac{L_3}{D_3^5} + \dots$$
- C) What is Siphon? Explain its working with neat sketch. **CO2 06**

*** End ***

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Winter End Semester Examination 2019

Course: Civil Engineering

Sem: III

Subject Name: Hydraulics I

Subject Code: BTCVC303

Max Marks: 60

Date: 14 /12/2019

Time: 10 AM to 1 PM

Duration: 3 Hr.

Instructions to the Students:

1. Solve ANY FIVE questions out of the following.
2. The level question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.
5. Solve All Questions in Sequential order.

	(CO)	Marks
Q. 1 Solve the following.	CO 2	12
A) Explain in detail Viscosity, its types and effect of temperature on Viscosity and solve below numerical: Two large Horizontal Plates of infinite extent are 2.5 cm apart and have liquid viscosity of $0.8 \text{ Ns} / \text{m}^2$ filled between them. What force is required to pull a thin plate of negligible thickness and area 0.5 m^2 between two large plates at a speed of 0.5 m/sec . if		06
a) The thin Plate is in the middle of the two large plates?		
B) Derive an Expression for Capillary rise in Glass Tube and solve below numerical: For having blood sample a fine glass capillary of dia. 2 mm was held touching a freshly punctured finger tip. Estimate in 'ml' the volume of blood sample so drawn. Take the density of blood as $1060 \text{ kg} / \text{m}^3$, surface tension as $5 \times 10^{-2} \text{ N/m}$ and its contact angle with glass as zero degrees.		06
Q.2 Solve the following.	CO 2	12
A) Explain and prove Pascal's Law. Also solve below numerical : A Triangular lamina of base and altitude equal to 3.2m is submerged in water vertically. Determine the Total pressure and centre of pressure, when Base of lamina coincides with free surface.		06
B) Derive an expression to find Metacentric Height of floating body by analytical method. Also solve below numerical: A wooden cylinder of diameter 'd' and length '2d' floats in water with its axis vertical. Is the equilibrium stable? Locate the meta-centre with reference to water surface. Specific gravity of wood is 0.60.		06

- Q. 3 Solve the following.** **CO 4 12**
- A) Derive Continuity Equation for three dimensional flows in Cartesian coordinate system. **06**
- B) Water flows through a horizontal pipe of diameter 20 cm with a velocity of 20 m/sec. The Pressure at A and B is given as 25 N/cm² and 20 N/cm² respectively. Find a head loss between A and B. **06**
- Q.4 Solve the following.** **CO 3 12**
- A) An Oil of Specific gravity 0.9 and Viscosity 0.12 Ns / m² flow through a pipe of 2.5 cm diameter with a pressure drop of 0.35 N / cm² in a length of 30m. Determine the discharge ,shear stress intensity at the pipe wall and the power required to maintain the flow. **06**
- B) Crude oil of specific gravity 0.9 and $\nu = 0.15 \text{ cm}^2/\text{sec}$ is flowing through a 25 cm diameter smooth pipe at the rate of 105 LPS. Calculate velocity at a distance of 6 cm from pipe axis, maximum velocity and velocity at the end of laminar sub layer. **06**
- Q. 5 Solve the following.** **CO 3 12**
- A) Define and explain below Dimensionless Numbers- a) Reynolds Number, b) Froude Number, c) Euler Number, d) Mach Number e) Weber Number **06**
- B) The Capillary rise ' h ' of a fluid of density ρ and surface tension σ in a tube of diameter D depends on contact angle θ and gravity g . Prove by Raleigh's method. **06**
- $$\frac{h}{D} = f_n \left[\frac{s}{\rho g D^2} \theta \right].$$
- Q. 6 Solve the following.** **CO 3 12**
- A) Two reservoirs 1 and 2 having level difference between free surfaces of water equal to 5m are connected by pipes AB and BC of diameters 20cm and 30cm respectively. Length of AB is 300m and that of BC is 200m.If flow through the pipeline is 50LPS, Find Loss due to expansion and friction factor f . Draw EG line. **06**
- B) A Siphon Pipe of 20cm diameter 500m long connects two tanks having difference of level in free surface of water of 20m.The summit is at 100 m length from upper tank and 3m above water level of upper tank. Find Q through siphon and pressure at summit .Take $f = 0.02$ **06**

*** End of the paper ***

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**End Semester Examination – Summer 2019****Course: B. Tech in Civil Engineering****Subject Name: Hydraulics - I****Max Marks:60****Date: 30-05-2019****Sem: III****Subject Code: BTCVC303****Duration: 3 Hr.****Instructions to the Students:**

1. Solve **ANY FIVE** questions out of the following.
2. The level question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1 Solve Any Two of the following.**12**

- A) Explain Newton's law of viscosity. Explain the importance of viscosity in fluid motion. (CO2)
Convert 1 kg/s-m dynamic viscosity in poise.
- B) The dynamic viscosity of oil, used for lubrication between a shaft and sleeve is 6 poise, the shaft is of diameter 0.4 m and rotates at 190 r.p.m. Calculate the power lost in the bearing for a sleeve length of 90mm. The thickness of oil film is 1.5mm. (CO2)
- C) The surface tension of water in contact with air at 20°C is 0.0725N/m. the pressure inside a droplet of water is to be 0.02N/cm² greater than outside pressure, calculate the diameter of droplet of water. (CO2)

Q.2 Solve Any Two of the following.**12**

- A) Derive an expression for the force exerted on a submerged vertical plane surface by the static liquid and locate the position of centre of pressure. (CO2)
- B) Define the following terms: (CO2)
- a. Buoyancy
 - b. Centre of Buoyancy
 - c. Meta –centre
- C) A rectangular plane surface 2m wide and 3m deep lies in water in such way that its plane makes an angle of 30° with the free surface of water. Determine the total pressure and position of centre of pressure when the upper edge is 1.5m below the free water surface. (CO2)

Q. 3 Solve Any One of the following.**12**

- A) 1. Define the terms : a) Velocity potential function, b) Stream function (CO1)
2. Define equation of continuity. Obtain an expression for continuity equation for three- dimensional flow.

- B)**
1. The diameter of pipe at the sections 1 and 2 are 10 cm and 15 cm respectively. Find the discharge through the pipe if the velocity of water flowing through the pipe at section 1 is 5m/s. Determine also the velocity at section 2. (CO1)
 2. Define the following terms:
 - a. Laminar flow
 - b. Turbulent flow
 - c. Steady flow

Q.4 Solve Any Two of the following.

12

- A)** Determine the dimensions of following terms
- a. Kinematic viscosity
 - b. Discharge
 - c. force
- (CO4)
- B)** What is meant by geometric, kinematic and dynamic similarities? (CO4)
- C)** The pressure drop in an aeroplane model of size 1/10 of its prototype is 80 N/cm². The model is tested in water. Find the corresponding pressure drop in prototype. Take density of air = 1.24 kg/m³. The viscosity of water is 0.01 poise while the viscosity of air is 0.00018 poise. (CO4)

Q. 5 Solve Any One of the following.

12

- A)** Derive an expression for the velocity distribution for laminar flow through a circular pipe. Also sketch the velocity distribution and shear stress distribution across a section of the pipe. (CO4)
- B)** A crude oil of viscosity 0.97 poise and relative density 0.9 poise and relative density 0.9 is flowing through a horizontal circular pipe of diameter 100mm and of length 10m. Calculate the difference of pressure at the two ends of the pipe, if 100kg of the oil is collected in a tank in 30 seconds. (CO4)

Q.6 Solve Any two of the following.

12

- A)** The difference in water surface level in two tanks, which are connected by three pipes in series of length 300m, 170m and 210m and of diameters 300mm, 200m and 400mm respectively, is 12m. Determine the rate of flow of water if coefficient of friction are 0.005, 0.0052 and 0.0048 respectively, considering a) minor losses (CO3)
- B)** In above question (Q.6 A) Determine the rate of flow of water by neglecting minor losses. (CO3)
- C)** Define the following terms: (CO3)
- a) Reynold's number, b) Froude's number, c) mach's number.

*** End ***

	DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Winter Examination – 2022 Course: B. Tech. Branch : Civil Engineering Semester : III Subject Code & Name: BTCVC304 Hydraulics-I Max Marks: 60 Date:15-03-2023 Duration: 3 Hr.			
	Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks	
Q. 1	Solve Any Two of the following.		12	
A)	Explain the different applications in Hydraulics Engineering	CO2	6	
B)	A Liquid has a Specific gravity of 1.9 and kinematic Viscosity of 6 stokes. What is its dynamic viscosity?	CO2	6	
C)	A rectangular plate 3m long and 1m wide is immersed vertically in water in such a Way that its 3m side is parallel to the water surface and is 1m below it. Find: i) Total Pressure on the plate ii) Position of Centre of Pressure	CO2	6	
Q.2	Solve Any Two of the following.		12	
A)	What is flow-net? Enumerate the methods of drawing flow nets. Also Define a) Velocity Potential Function b) Stream Function.	CO1	6	
B)	List the Flow Measuring Devices and Explain with neat Sketch U-Tube Differential Manometer	CO1	6	
C)	The water is flowing through a tapering pipe having diameters 300mm and 150mm at sections 1 and 2 respectively. The discharge through the pipe is 40litre/sec. The section is 10m above datum and section2 is 6m above datum. Find the intensity of Pressure at section2 if that a section1 is 400KN/m ² .	CO2	6	

Q. 3	Solve Any Two of the following.		12
A)	What do you meant by Prandtl mixing length theory? Find an expression for shear stress due to Prandtl.	CO4	6
B)	Derive an expression for Euler's equation of motion.	CO4	6
C)	Define Boundary layer and describe effect of pressure gradient on boundary layer separation with neat sketch	CO4	6

Q.4	Solve Any Two of the following.		12
A)	State Buckingham's - theorem. What do you mean by repeating variables? How are the repeating variables selected in dimensional analysis?	CO4	6
B)	Explain the concept of Water Hammer and write a note on Surge Tank	CO3	6
C)	Write short note on Non-dimensional numbers- i) Froude Number ii) Weber Number iii) Reynolds Number	CO4	6

Q. 5	Solve Any Two of the following.		12
A)	Explain With neat sketch Hydraulic gradient and Total energy Line	CO2	6
B)	What is Compound Pipe? What will be loss of head when pipes are connected in series?	CO3	6
C)	The rate of flow of water through a horizontal pipe is $0.25\text{m}^3/\text{s}$. The diameter of the pipe which is 200mm is suddenly enlarged to 400mm. The pressure intensity in the smaller pipe is $11.772 \times 10^4 \text{N/m}^2$. Determine i) loss of head due to sudden enlargement ii) Pressure intensity in larger pipe	CO3	6
	*** End ***		

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Supplementary Examination – Summer 2022

Course :B.Tech.

Branch : Civil Engg.

Semester : III

Subject Code & Name : BTCVC304 Hydraulics-1

Max Marks : 60

Date : / /

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q.1 Solve Any Two of the Following		
A) Explain the terms specific volume, specific gravity and density.	CO2	06
B) State the newton's law of viscosity and give examples of its applications.	CO2	06
C) Calculate the density, specific weight and weight of one liter of petrol of specific gravity =0.7	CO2	06
Q.2 Solve Any Two of the Following		
A) Explain flow net and its uses?	CO2	06
B) Derive Euler's equation of motion?	CO2	06
C) A 30 cm diameter pipe conveying water, branches into two pipe of diameter 20 cm and 15 cm respectively. If the average velocity in the 30 cm diameter pipe is 2.5 m/s. Find the discharge in this pipe, also determine the velocity in the 15 cm pipe if the average velocity in a 20 cm diameter pipe is 2 m/s?	CO3	06
Q.3 Solve Any Two of the Following		
A) Derive fully developed laminar flow between infinite parallel plate when both plates are stationary?	CO1	06
B) Explain Prandtl mixing length theory?	CO2	06
C) What is the boundary layer? Explain in detail.	CO2	06
Q.4 Solve Any two of the Following		
A) Derive Darcy-Weisbach equation.	CO2	06
B) Explain the terms prototype, model analysis and hydraulic similitude.	CO4	06
C) State and explain Buckingham's Pi theorem.	CO2	06

Q.5 Solve Any Two of the Following

- | | | | |
|----|--|-----|----|
| A) | What is major loss in flow through pipe and what causes the major loss ?
Derive the equation for the major loss. | CO3 | 06 |
| B) | Three pipes of lengths 800m, 500m and 400m and of diameters 500 mm, 400mm and 300mm respectively are connected in series. These pipes are to be replaced by single pipes of length 1700 m. Find the diameter of the single pipe. | CO3 | 06 |
| C) | Define and explain the terms : (i) Hydraulic gradient line and (ii) Total energy line | CO1 | 06 |

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Summer Examination 2023

Course: Second Year B. Tech.

Branch: Civil Engineering

Subject Name: Hydraulics - I

Subject Code: BTCVC304

Max Marks: 60

Date: 18/08/2023

Duration: 3 Hrs.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1 Solve Any Two of the following.

- A) Explain the following fluid properties: (i) Weight density, (ii) Specific gravity, (iii) viscosity. **Remember** **06**
- B) Enlist different types of manometers. Explain Differential U-Tube Manometer with neat sketch. **CO1** **06**
- C) A plate 0.0254 mm distant from a fixed plate, moves at 61 cm/sec and requires a force of $0.2 \text{ kg(f)}/\text{m}^2$ to maintain this speed. Determine the dynamic viscosity of the fluid between the plates. **CO2** **06**

Q. 2 Solve Any Two of the following.

- A) Derive a three dimensional general continuity equation in Cartesian co-ordinates. **Remember** **06**
- B) Enlist the methods of drawing flow nets. Describe the uses and limitations of flow net. **CO2** **06**
- C) The stream function for a two-dimensional flow is given by $\Psi = 2xy$, calculate the velocity at the point P (2,3). Find the velocity potential function Φ . **CO 2** **06**

Q. 3 Solve Any Two of the following.

- A) Derive an expression for velocity distribution for laminar flow between two fixed parallel plates. **Remember** **06**
- B) What is Prandtl mixing length theory? Find an expression for shear stress due to Prandtl. **CO2** **06**
- C) Write a short note on Boundary Layer Theory. **CO2** **06**

Q. 4 Solve Any Two of the following.

- A) Write short note on non-dimensional numbers – (i) Froude number, (ii) Weber Number, (iii) Reynold's Number, (iv) Euler Number, (v) Mach Number. **CO 3** **06**
- B) What is meant by geometric, kinematic and dynamic similarities? **CO 4** **06**

- C) Explain concept of Rayleigh's method and find an expression for the drag force using Rayleigh's method on smooth sphere of diameter D , moving with uniform velocity V in a fluid of density ρ and dynamic viscosity μ . **CO 4** **06**

Q. 5 Solve Any Two of the following.

- A) What is Siphon? Explain its working with neat sketch. **Remember** **06**
- B) What is a surge tank and what are its functions? Describe with neat sketches different types of surge tanks. **CO 4** **06**
- C) Find the diameter of a pipe of length 2000 m when the rate of flow of water through the pipe is 200 *liters/s* and head lost due to friction is 4 m. Take the value of $C = 50$ in *Chezy's formulae*. **CO 3** **06**

----- **End** -----

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular & Supplementary Winter Examination - 2023

Course: Second Year B. Tech. (Sem III)

Branch: Civil Engineering

Subject Name: Hydraulics-I

Subject Code: BTCVC304

Max Marks: 60

Duration: 3 Hrs.

Date: 09/01/2024

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) (1) Explain the phenomenon of the capillarity. Obtain the expression of capillary rise of a liquid.	CO2	6
(2) At a certain point in castor oil the shear stress is 0.432 N/m^2 and the velocity gradient 0.216 s^{-1} . If the mass density of castor oil is 959.42 kg/m^3 , find kinematic viscosity.		
B) An inverted U-tube differential manometer is connected to two horizontal pipes A and B through which water is flowing. The vertical distance between the axes of these pipes is 30 cm. When an oil of specific gravity 0.8 is used as a gauge fluid, the vertical heights of water columns in the two limbs of the inverted manometer (when measured from the respective centre of the pipe) are found to be same i.e. 35cm. Determine the pressure between the pipe.	CO2	6
C) Define buoyancy and centre of buoyancy. Derive an expression for the metacentric height of a floating body.	CO2	6
Q.2 Solve Any Two of the following.		12
A) A 30 cm diameter pipe, conveying water, branches into two pipes of diameters 20 cm and 15 cm respectively. If the average velocity in 30 cm diameter pipe is 2.5 m/s, find the discharge in this pipe. Also determine the velocity in 15 cm pipe if the average velocity in 20 cm diameter pipe is 2 m/s.	CO1	6
B) The velocity potential function is given by an expression of $\phi = -\frac{xy^3}{3} - x^2 + \frac{x^3y}{3} + y^2$ a. Find the velocity components in x and y directions respectively. b. Show that ϕ represents a possible case of flow.	CO1	6
C) Derive Euler's equation of motion. Explain how this is integrated to get Bernoulli's equation.	CO1	6

Q. 3 Solve Any Two of the following.		12
A) Derive the expression for shear stress distribution and velocity distribution in fully developed laminar flow through pipe.	CO4	6
B) Write a short note on Boundary Layer Theory.	CO4	6
C) What is Prandtl mixing length theory? Find an expression for shear stress due to Prandtl.	CO4	6
Q.4 Solve Any Two of the following.		12
A) Explain concept of Rayleigh's method and find an expression for the time period using Rayleigh's method, where time period (t) of a pendulum depends upon the length (L) of the pendulum and acceleration due to gravity (g).	CO4	6
B) Give short explanation about following non-dimensional numbers – (i) Froude number, (ii) Euler Number, (iii) Reynold's Number, (iv) Mach number, (v) Weber number	CO4	6
C) Compare between distorted and undistorted models with the suitable example.	CO4	6
Q. 5 Solve Any Two of the following.		12
A) What is Compound Pipe? What will be loss of head when pipes are connected in series?	CO3	6
B) Describe Water Hammer and write a note on Surge Tank.	CO3	6
C) What are the uses of Nomogram and Moody's chart?	CO3	6

***** End *****

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Summer Examination – 2024

Course: B. Tech.

Branch: Civil Engineering

Semester: III

Subject Code & Name: BTCVC304

Hydraulics-I

Max Marks: 60

Date: 06/07/2024

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) Define following fluid properties with short explanation:- specific weight, specific volume, specific gravity, viscosity, surface tension, capillarity.	CO2	6
B) State and prove Pascal's law with neat diagram.	CO2	6
C) Define buoyancy and metacentric height. Derive an expression for the metacentric height of a floating body.	CO2	6
Q.2 Solve Any Two of the following.		12
A) The velocity components in a two-dimensional flow field for an incompressible fluid are expressed as $u = 2x + \frac{y^3}{3} - yx^2 ; v = -2y - \frac{x^3}{3} + xy^2$	CO1	6
(a) Obtain an expression for stream function Ψ		
(b) Obtain an expression for velocity potential Φ		
B) Derive Euler's equation of motion. Explain how this is integrated to get Bernoulli's equation.	CO1	6
C) Discuss briefly the different methods of drawing the flow net.	CO1	6
Q. 3 Solve Any Two of the following.		12
A) Derive an expression for mean velocity for laminar flow (i) through a pipe, (ii) between two parallel plates-both plates at rest.	CO4	6
B) Write a short note on Nikuradse's experiment.	CO4	6
C) Explain the boundary layer theory.	CO4	6
Q.4 Solve Any Two of the following.		12
A) Explain methods of dimensional analysis.	CO4	6

- | | | |
|---|-----|---|
| B) Explain different types of similarities that must exist between a prototype and its model. | CO4 | 6 |
| C) Explain the terms: distorted models and undistorted models. What is the use of distorted models? | CO4 | 6 |

Q. 5 Solve Any Two of the following. **12**

- | | | |
|--|-----|---|
| A) What is syphon? Explain on what principle it works? | CO3 | 6 |
| B) Discuss the concept of Water Hammer and Surge Tank. | CO3 | 6 |
| C) Explain the Nomogram and Moody's chart. | CO3 | 6 |

***** End *****